

Adriana-Stefania Ciupeanu

Manitoba, Canada · [GitHub](#) · [Website](#)

RESEARCH PROFILE

Statistician and mathematical modeller developing Bayesian, machine-learning, and network-based methods for complex biological and dynamical systems. Research interests include Bayesian inference, uncertainty quantification, neural surrogate modelling, temporal and spatial networks, and reproducible scientific software. Current research focuses on combining mechanistic modelling with modern statistical learning, with applications in infectious-disease dynamics, population health, and interconnected spatial systems.

ACADEMIC APPOINTMENTS

Postdoctoral Fellow

May – October 2026

Department of Mathematics, University of Manitoba, Winnipeg, Manitoba, Canada

Supervisor: Dr. Julien Arino

- Lead the development of a machine-learning extension to an existing global metapopulation model for forecasting the spatiotemporal risk of pathogen importation across 238 countries and territories connected through air- and ground-transportation networks.
- Conceived and developed a mechanistically informed neural-surrogate methodology to evaluate whether machine-learning approximations can make repeated evaluation of a large, coupled ODE model tractable for likelihood-free Bayesian inference and probabilistic forecasting.
- Designed and implemented temporally conditioned neural-surrogate architectures, together with comparative spatiotemporal graph attention and recurrent neural-network models incorporating transportation-network and location-level features.
- Developed an end-to-end Python research pipeline encompassing mechanistic simulation, parameter-space sampling, surrogate training, SMC-ABC inference, posterior uncertainty propagation, validation against observed first-case records, and computational benchmarking.

EDUCATION

Ph.D. Interdisciplinary Individual Studies (IIS) — Mathematics and Statistics 2019–2026

University of Manitoba, Winnipeg, Manitoba, Canada

- Supervisors: Dr. Julien Arino and Dr. Saman Muthukumarana
- Dissertation Title: Mathematical and statistical modelling during a pandemic event

Master of Science — Mathematics

2016 – 2020

University of Manitoba, Winnipeg, Manitoba, Canada

Master of Science (Course-Based) — Mathematics and Statistics

2015 – 2016

Queen's University, Kingston, Ontario, Canada

Bachelor of Science (Honours) — Mathematics

2012 – 2015

Oxford Brookes University, Oxford, Oxfordshire, UK

RESEARCH INTERESTS

- Statistical inference and computational methods for mechanistic models of complex biological and health systems

- Bayesian computation, parameter identifiability, uncertainty quantification, and global sensitivity analysis
- Machine learning for structured, spatial, temporal, and network data, including graph neural networks and neural surrogate models
- Statistical and machine-learning methods for partially observed systems and heterogeneous data integration
- Reproducible computational research and open-source statistical software development

TECHNICAL SKILLS

- **Programming:** Python, R, Git, GitHub, L^AT_EX
- **Computing:** PyTorch, PyTorch Geometric, NumPy, SciPy, pandas, scikit-learn, NetworkX, igraph
- **Statistical methods:** Bayesian MCMC, SMC-ABC, parameter estimation, identifiability analysis, uncertainty quantification, model comparison, PRCC, eFAST
- **Machine learning:** graph attention networks, recurrent neural networks (LSTM/GRU), neural surrogate modelling, mechanistically informed machine learning
- **Data and networks:** temporal and spatial network analysis, graph analysis, heterogeneous observational data, scientific NLP workflows
- **Research software:** Python package development, automated testing, documentation, version control, and reproducible computational workflows

PUBLICATIONS AND RESEARCH OUTPUTS

Refereed Journal Publications

1. **Ciupeanu, A.-S.**, Varughese, M., Roda, W.C., Han, D., Cheng, Q., Li, M.Y. (2022). Mathematical Modeling of the Dynamics of COVID-19 Variants of Concern: Asymptotic and Finite-Time Perspectives. *Infectious Disease Modelling*.

Manuscripts Submitted / Under Review

1. Akhter, T., **Ciupeanu, A.-S.**, Roda, W.C., Li, M.Y., Plitt, S.S., Charlton, C.L. Preventing HPV-Induced Cervical Cancer in Alberta, Canada. Submitted to *Canadian Journal of Public Health*.

Preprints

1. **Ciupeanu, A.-S.**, Wickramasinghe, A., Muthukumarana, S., Arino, J. Investigating Air Travel Network Changes. Preprint available online.

Manuscripts in Preparation

1. **Ciupeanu, A.-S.**, Varughese, M., Roda, W.C., Han, D., Wang, X., Akhter, T., Li, M.Y. Quantifying the Effects of Public Health Interventions in Alberta during the First Wave of COVID-19.
2. **Ciupeanu, A.-S.**, Arino, J., Portet, S. Neural-Surrogate Modelling for Forecasting of Pathogen Importation Risk. Manuscript in preparation.

Research Software

1. **Ciupeanu, A.-S.**, Arino, J. `temporal_networks`: Python package for temporal network analysis. Lead developer; preparing for submission to pyOpenSci.
2. Arino, J., **Ciupeanu, A.-S.** `wikipediaGATN`: Python package for constructing global air-transportation networks from public data. Submitted to pyOpenSci for peer review.

Technical and Government Reports

1. Arino, J., Portet, S., Bajeux, N., **Ciupeanu, A.-S.** Investigation of Global and Local COVID-19 Importation Risks. Report to the Public Health Risk Science Division, Public Health Agency of Canada.

TEACHING EXPERIENCE

Instructor — Mathematical Skills (MSKL 0100)

Fall 2025

Extended Education, University of Manitoba

- *Level: Foundational / pre-university mathematics — Instruction: Lectures — Independence: Sole instructor*
- Held full responsibility for course delivery, assessment and examination design, grading, learning materials, feedback, and student support.
- Nominated for the Extended Education Instructor Excellence Award, University of Manitoba (2026).

Lead Teaching Assistant & Guest Lecturer — Mathematics of Data Science (MATH 2740)

2023–2025

University of Manitoba

- *Level: Undergraduate — Instruction: Lectures and laboratory support*
- Delivered three guest lectures on graph theory and network science, independently developing lecture notes and teaching materials from assigned topics.
- Served as Lead Teaching Assistant, with responsibility for grading programming assignments and contributing datasets and revisions to course assessments.

Workshop Instructor — Introduction to Calculus (MATH 1500)

Fall 2025

University of Manitoba

- *Level: First-year undergraduate — Instruction: Weekly 2-hour supplementary workshops — Independence: Independently designed and delivered*
- Designed and delivered weekly workshops, typically attended by 70–90 students, combining interactive instruction and worked examples with independently developed practice worksheets.
- Adapted instruction and pre-examination review sessions in response to recurring difficulties identified through student questions and problem solving.

Laboratory Instructor — Basic Statistical Analysis (STA 1000)

2026

International College of Manitoba, Winnipeg, Manitoba, Canada

- *Level: First-year undergraduate — Instruction: Statistics and R laboratory*
- Introduced students with no prior R experience to statistical computing, including data manipulation and visualisation, descriptive statistics, confidence intervals, hypothesis testing, and regression.
- Supplemented instructor-provided materials with additional code and explanations, and provided individual support with implementation and debugging.

Laboratory Coordinator — Calculus 2 (MATH 1700)

Fall 2023, Winter 2026

University of Manitoba

- Coordinated laboratory instruction and assessment across multiple sections and served as the primary point of contact for teaching assistants.
- Developed five equivalent versions of each laboratory quiz from an instructor-provided version, prepared solutions, and managed administration through Crowdmark.
- Coordinated grading and student accommodations relating to laboratory assessments.

Laboratory Coordinator — Vector Geometry and Linear Algebra (MATH 1300) Winter 2026

University of Manitoba

- Coordinated laboratory instruction and assessment across multiple sections and served as the primary point of contact for teaching assistants.
- Prepared section-specific quizzes from instructor-provided question sets and coordinated administration and grading through Crowdmark.

Additional Teaching Assistant Experience

2016–2025

University of Manitoba

Delivered laboratory/tutorial instruction, grading, examination proctoring, and individual student support across undergraduate mathematics and data-science courses. Earlier calculus appointments included autonomy in structuring laboratory sessions around instructor-specified topics.

- **Pre-Calculus in Practice (MATH 1018)** — 2025
- **Vector Geometry and Linear Algebra (MATH 1300)** — 2023–2024
- **Techniques of Classical and Linear Algebra (MATH 1210)** — 2019–2020
- **Elementary Discrete Mathematics (MATH 1240)** — 2018, 2024
- **Introduction to Calculus (MATH 1500)** — multiple sections, 2016–2024
- **Introductory Calculus for Management and Social Sciences (MATH 1520)** — 2019, 2023
- **Linear Algebra 2 (MATH 2090)** — 2017
- **Modelling (MATH 2140)** — 2018–2019
- **Mathematics of Data Science (MATH 2740)** — 2023–2025

Additional Laboratory Instruction

2018–2026

International College of Manitoba, Winnipeg, Manitoba, Canada

Delivered laboratory instruction in a university pathway programme serving predominantly international students with varied educational backgrounds. Appointments included independently designed laboratories as well as facilitation of instructor-provided materials.

- **Applied Finite Mathematics (MAT 1010)** — 2018–2019, 2024, 2026; independently designed laboratories in 2018–2019
- **Techniques of Classical and Linear Algebra (MAT 1210)** — 2025–2026; independently designed laboratory sessions
- **Algebra / Vector Geometry and Linear Algebra (MAT 1300)** — 2018–2025
- **Elementary Discrete Mathematics (MAT 1240)** — 2023–2026
- **Mathematics for Management and Social Sciences (MAT 1524)** — 2024–2026

RESEARCH SUPERVISION AND MENTORING

Undergraduate Research Mentor

2026–present

University of Manitoba

- Provide day-to-day research mentoring for an undergraduate project developing a hybrid gravity–graph neural network model of inter-city mobility for spatial epidemic modelling; formal faculty supervisor: Dr. Julien Arino.
- Provide methodological direction on graph neural networks, mobility modelling, and metapopulation epidemic models, helping refine the student’s initial project scope into a feasible research and computational framework.
- Meet regularly with the student to guide model implementation, debugging, experimental design, validation, interpretation of results, and research communication.

Undergraduate Research Mentor

Summer 2023

University of Manitoba

- Mentored an undergraduate researcher analysing Alberta COVID-19 behavioural survey data, providing statistical and computational guidance in R.
- Helped develop a behavioural-compliance framework based on masking and social-distancing responses and guided the use of survey-weighted analyses to compare compliance across epidemic waves and demographic groups.
- Advised on data cleaning and recoding, weighted descriptive summaries, cross-tabulations, and multinomial logistic regression examining associations between compliance level, age, sex, geographic zone, and epidemic wave.
- Met weekly with the student and faculty supervisor to review results, propose next analytical steps, troubleshoot R code, and support interpretation of statistical findings.

TALKS AND POSTER PRESENTATIONS

Conference Presentations

- Preventing HPV-Induced Cervical Cancer in Alberta, Canada: A Mathematical Modelling Study — CMS Winter Meeting, Vancouver, Canada, December 2024
- Dynamics of COVID-19 Variants of Concern — CMS Winter Meeting, Montreal, Canada, December 2023
- Dynamics of COVID-19 Variants of Concern — CMPD 6, Winnipeg, Canada, May 2023
- Changes in Air Travel Patterns during COVID-19 — PRIMA Congress, Vancouver, Canada, December 2022
- Changes in Air Travel Patterns during COVID-19 — Mathematics for Public Health Festival (MfPHest), Toronto, Canada, October 2022
- Bayesian Inference and COVID-19 — CAIMS, Remote, July 2021
- Addressing the Scarcity of Data during Epidemic Events — CDM Incubation Day, Remote, February 2021

Seminars

- Mathematical Modeling and Data Science Approaches for COVID-19 Epidemiology — Math Bio Seminar, University of Manitoba, September 2025

CONFERENCES ORGANIZED OR CO-ORGANIZED

- CDM Incubation Day (co-organizer) February 2021

AWARDS, FUNDING AND RECOGNITION

- Nominated for the Extended Education Instructor Excellence Award, University of Manitoba (2026)
- Faculty of Graduate Studies Travel Award, University of Manitoba 2022–2023, 2023–2024
- Graduate Enhancement of Tri-Agency Stipends Program 2016, 2017, 2021, 2022, 2023
- Visual and Automated Disease Analytics (VADA) Graduate Training Program. NSERC CREATE-funded interdisciplinary training in infectious-disease analytics, statistical modelling, machine learning, and health data science 2019–2021
- International Graduate Student Entrance Scholarship, University of Manitoba 2019–2020
- International Graduate Student Scholarship, University of Manitoba 2017–2018
- International Graduate Student Entrance Scholarship, University of Manitoba 2016–2017
- Dr. Narain D. Gupta Scholarship in Mathematics, University of Manitoba 2016–2017
- Queen's Graduate Awards, Queen's University 2015–2016
- Tuition Award, Queen's University 2015–2016
- IMA Prize for Academic Excellence in Mathematics, Oxford Brookes University 2015
- Department of Mechanical Engineering and Mathematical Sciences Student Prize, Oxford Brookes University 2015

ACADEMIC SERVICE

Graduate Student Representative, Faculty Hiring Committee

2024–2025

Department of Mathematics, University of Manitoba

- Served as the graduate student representative throughout a faculty recruitment process, contributing to the development of the position advertisement, selection criteria, and interview questions.
- Reviewed and scored applications using an agreed assessment rubric and participated in candidate shortlisting and evaluation.
- Participated in candidate interviews and campus visits, meetings with shortlisted candidates, and committee deliberations leading to the hiring recommendation.

Other Academic Services

- Graduate Student Representative, Department of Mathematics, University of Manitoba
2018–2019
- Graduate Student Representative, Department Councillor, University of Manitoba Graduate Student Association
2018–2019
- Committee Member, Graduate Student Association
2018–2019
- Production/Layout Editor, Proceedings of Manitoba's Undergraduate Science and Engineering Research
2018

REFERENCES

- **Julien Arino**
Professor, Department of Mathematics, University of Manitoba
- **Stéphanie Portet**
Professor, Department of Mathematics, University of Manitoba
- **Saman Muthukumarana**
Professor and Department Head, Statistics, University of Manitoba
- **Michael Y. Li**
Professor, Department of Mathematical and Statistical Sciences, University of Alberta